Example 6.10

A particle is acted upon by the forces $3\hat{i}-2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$ is displaced from the point $(1,3,-1)$ to the point $(4,-1,\lambda)$. If the work done by the forces is 16 units, find the value of λ.

Solution

Resultant of the given forces is $\vec{F}=(3\hat{i}-2\hat{j}+2\hat{k})+(2\hat{i}+\hat{j}-\hat{k})=5\hat{i}-\hat{j}+\hat{k}$.

The displacement of the particle is given by

$$\vec{d}=(4\hat{i}-\hat{j}+\lambda\hat{k})-(\hat{i}+3\hat{j}-\hat{k})=(3\hat{i}-4\hat{j}+(\lambda+1)\hat{k}).$$

As the work done by the forces is 16 units, we have

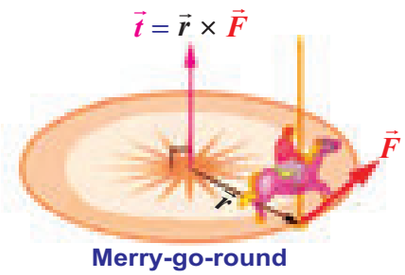
$$\vec{F}\cdot\vec{d}=16.$$

That is, $(5\hat{i}-\hat{j}+\hat{k})\cdot(3\hat{i}-4\hat{j}+(\lambda+1)\hat{k}=16\Rightarrow\lambda+20=16$.

So, $\lambda=-4$.

Definition 6.3

If a force $\vec{F}$ is applied on a particle at a point with position vector $\vec{r}$, then the **torque** or **moment** on the particle is given by $\vec{t}=\vec{r}\times\vec{F}$. The torque is also called the rotational force.



Merry-go-round

Fig. 6.15

Example 6.11

Find the magnitude and the direction cosines of the torque about the point $(2,0,-1)$ of a force $2\hat{i}+\hat{j}-\hat{k}$, whose line of action passes through the origin.

Solution

Let A be the point $(2,0,-1)$. Then the position vector of A is $\overrightarrow{OA}=2\hat{i}-\hat{k}$ and therefore $\vec{r}=\overrightarrow{AO}=-2\hat{i}+\hat{k}$.

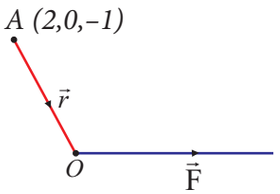


Fig. 6.15

Then the given force is $\vec{F}=2\hat{i}+\hat{j}-\hat{k}$. So, the torque is

$$\vec{t}=\vec{r}\times\vec{F}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&0&1\\2&1&-1\end{vmatrix}=-\hat{i}-2\hat{k}.$$

The magnitude of the torque $=|-\hat{i}-2\hat{k}|=\sqrt{5}$ and the direction cosines of the torque are $-\frac{1}{\sqrt{5}},0,-\frac{2}{\sqrt{5}}$.

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